

Q1. A small business has purchased a new **Cisco network switch** to set up its office network. The network administrator needs to perform a **basic switch configuration**, including **setting up hostname, passwords, VLAN configuration and configuring port security** to prevent unauthorized access. Once configured, the setup is tested by **checking switch status using basic commands**.

Question:

"How can we configure a basic switch setup on a Cisco network switch to enable secure and efficient network operation?"

Q2. A university's **IT department** is setting up VLANs for different faculties (**CSE and ECE**) to improve network segmentation. To enable inter-switch (2960) communication, **trunking** is configured. The setup is simulated in **Cisco Packet Tracer**, and connectivity is verified using **ping tests and VLAN/trunk status checks**.

Question:

"How can we configure multiple VLANs and implement trunking in Cisco Packet Tracer for efficient network segmentation?"

Q3. A company has two departments (**Sales, HR**) connected through a **Cisco switch (2960)**. To allow communication between these departments while maintaining network segmentation, the administrator configures **Inter-VLAN routing** using **802.1Q encapsulation** on a router-on-a-stick setup in **Cisco Packet Tracer**. After configuration, connectivity is verified by **pinging devices across VLANs and checking the routing table**.

Question:

"How can we configure 802.1Q encapsulation in Cisco Packet Tracer to enable inter-VLAN communication using a router-on-a-stick setup?"

Q4. A homeowner wants to automate their house using a **smart home network** in **Cisco Packet Tracer**. The setup includes **smart lights, smart doors, and temperature sensors**, all connected to a **home gateway/router** for centralized control. The administrator configures the network to enable remote access and automation. After configuration, the system is tested by **turning devices on/off, verifying network connectivity, and ensuring proper communication between smart devices and the home gateway**.

Question:

"How can we implement a smart home network using Cisco Packet Tracer and verify its configuration for seamless automation and control?"

Q5. A company has two separate office branches, each with its own network (**192.168.1.0/24 for Sales and 192.168.2.0/24 for HR**). To enable communication between these networks, Routers are used to configure **static routing**. Additionally, the administrator checks the **ARP table** to ensure proper MAC address resolution for connected devices. After configuration, connectivity is verified using **ping tests, ARP table inspection, and route table checks**.

Question:

"How can we configure static routing on a Cisco network router, check the ARP table, and verify connectivity between different networks?"

Q6. A company has two separate office branches, each with its own network (**192.168.1.0/24 for Sales and 192.168.2.0/24 for HR**). Since the number of users in each branch is very small, the communication between these networks uses **RIP (Routing Information Protocol)**. After configuration, connectivity is verified by **checking routing tables and using ping/traceroute commands to ensure successful communication between networks**.

Question:

"How can we configure RIP on a Cisco network router and verify connectivity between multiple branch networks?"

Q7. A company has two separate large office branches, each with its own network (**192.168.1.0/24 for Sales and 192.168.2.0/24 for HR**). **OSPF** is implemented for **scalability and faster convergence** in larger offices. After configuration, connectivity is verified by **checking routing tables and using ping/traceroute commands to ensure successful communication between networks**.

Question:

"How can we configure OSPF on a Cisco network router and verify connectivity between multiple branch networks?"

Q8. Configuration of Network address translation in Cisco packet tracer and verify the configuration

A small business has a **private network (10.0.0.0/8)** and needs to allow its internal devices to access the internet using a single **public IP address (200.100.0.1)**. The network administrator configures **Network Address Translation (NAT)** on a **Cisco router in Cisco Packet Tracer** to enable this communication. After configuration, connectivity is verified by **checking NAT translation tables and using ping/traceroute commands to confirm successful address translation**.

Question:

"How can we configure Network Address Translation (NAT) in Cisco Packet Tracer and verify its functionality for enabling internet access from a private network?"